# Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

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**Ringkasan**

PENDAHULUAN
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[utf8]inputenc amssymb amsfonts amsmath bigints arcsec arccot arccsc  100 Integrals Josiah Rondaris

## 1 Basic Integrals to Remember

$\int x^n dx = \frac{1}{n+1} x^{n+1} + C$
$\int \sin(x)\, dx = -\cos(x) + C$
$\int \cos(x)\, dx = \sin(x) + C$
$\int \tan(x)\, dx = -\ln|\cos(x)| + C$ or $\ln|\sec(x)| + C$
$\int \csc(x)\, dx = -\ln|\csc(x) + \cot(x)| + C$
$\int \sec(x)\, dx = \ln|\sec(x) + \tan(x)| + C$
$\int \cot(x)\, dx = \ln|\sin(x)| + C$ or $-\ln|\csc(x)| + C$
$\int a^x dx = \frac{1}{\ln(a)} a^x + C$
$\int e^x dx = e^x + C$
$\int \log_a(x)\, dx = x\log_a(x) - \frac{1}{\ln(a)} x + C$
$\int \ln(x)\, dx = x\ln(x) - x + C$

## 2 Advanced Integrals to Remember

$\int \sec^3(x) dx = \frac{1}{2}\sec(x)\tan(x) + \frac{1}{2}\ln|\sec(x) + \tan(x)| + C$
$\int \arcsin(x) dx = x\arcsin(x) + \sqrt{1-x^2} + C$
$\int \arccos(x) dx = x\arccos(x) - \sqrt{1-x^2} + C$
$\int \arctan(x) dx = x\arctan(x) - \frac{1}{2}\ln(x^2+1) + C$
$\int \sqrt{x^2+a^2} dx = \frac{1}{2}x\sqrt{x^2+a^2} + \frac{a^2}{2}\ln\left|\sqrt{x^2+a^2} + x\right| + C$
$\int \sqrt{x^2-a^2} dx = \frac{1}{2}x\sqrt{x^2-a^2} - \frac{a^2}{2}\ln\left|\sqrt{x^2-a^2} + x\right| + C$
$\int \sqrt{a^2-x^2} dx = \frac{a^2}{2}\arcsin\left(\frac{x}{a}\right) + \frac{1}{2}x\sqrt{a^2-x^2} + C$

$$\int \frac{1}{x^2+a^2}dx = \frac{1}{a}\arctan\left(\frac{x}{a}\right) + C$$
$$\int \frac{1}{x^2-a^2}dx = \frac{1}{2a}\ln\left|\frac{x-a}{x+a}\right| + C$$
$$\int \frac{1}{a^2-x^2}dx = \frac{1}{2a}\ln\left|\frac{a+x}{a-x}\right| + C$$
$$\int \frac{1}{\sqrt{x^2+a^2}}dx = \ln\left|x+\sqrt{x^2+a^2}\right| + C$$
$$\int \frac{1}{\sqrt{x^2-a^2}}dx = \ln\left|x+\sqrt{x^2-a^2}\right| + C$$
$$\int \frac{1}{\sqrt{a^2-x^2}}dx = \arcsin\left(\frac{x}{a}\right) + C$$
$$\int \sin^n(x)dx = -\frac{1}{n}\sin^{n-1}(x)\cos(x) + \frac{n-1}{n}\int \sin^{n-2}(x)dx$$
$$\int \cos^n(x)dx = \frac{1}{n}\cos^{n-1}(x)\sin(x) + \frac{n-1}{n}\int \cos^{n-2}(x)dx$$

# 3 The 100 Integrals

## Problem 1

$$\int \frac{1}{\sqrt{x}\,(1+x)}dx$$

Substituting u=$\sqrt{x}$
$du = \frac{1}{2\sqrt{x}}dx$ yields:

$$= 2\int \frac{1}{1+u^2}du$$

This integral is standard:

$$= 2\arctan(u) + C$$

Undoing the substitution(s):

$$= 2\arctan\left(\sqrt{x}\right) + C$$

## Problem 2

$$\int \frac{\sec^2(x)}{1+\tan(x)}dx$$

Substituting u=1+tan($x$)
$du = \sec^2(x)dx$ yields:

$$= \int \frac{1}{u}du$$

This integral is standard:

$$= \ln|u| + C$$

Undoing the substitution(s):

$$= \ln|1+\tan(x)| + C$$

## Problem 3

$$\int \sin(x)\sec(x)\,dx$$

Rewriting the integral:

$$= \int \frac{\sin(x)}{\cos(x)}dx$$

Substituting u=cos($x$)
$du = -\sin(x)dx$ yields:

$$= -\int \frac{1}{u}du$$

This integral is standard:

$$= -\int \ln|u| + C$$

Undoing the substitution(s):

$$= -\ln|\cos(x)| + C$$

## Problem 4

$$\int \frac{\csc(x)\cot(x)}{1+\csc^2(x)}dx$$

Substituting u=csc($x$)
$du = -\csc(x)\cot(x)dx$ yields:

$$= -\int \frac{1}{1+u^2}du$$

This integral is standard:

$$= -\arctan(u) + C$$

Undoing the substitution(s):

$$= -\arctan(\csc(x)) + C$$

## Problem 5

$$\int \frac{\tan(x)}{\cos^2(x)}dx$$

Rewriting the integral:

$$= \int \tan(x)\sec^2(x)dx$$

Substituting u=tan($x$)
$du = \sec^2(x)dx$ Yields:

$$= \int u du$$

This integral is standard:

$$= \frac{1}{2}u^2 + C$$

Undoing the substitution(s):

$$= \frac{1}{2}\tan^2(x) + C$$

## Problem 6

$$\int \csc^4(x)\,dx$$

Rewriting the integral: $=\int \csc^2(x)\cdot\csc^2(x)dx$
$= \int \csc^2(x)(\cot^2(x)+1)dx$ Substituting u=cot($x$)
$du = -\csc^2(x)dx$ Yields: $=-\int(u^2+1)du$
$= -\int u^2du - \int du$ Both integrals are standard:

$$= -\frac{1}{3}u^3 - u + C$$

Undoing the substitution(s):

$$= -\frac{1}{3}\cot^3(x) - \cot(x) + C$$

## Problem 7

$$\int x \tan^2(x)\, dx$$

Rewriting the integral: $=\int x(\sec^2(x)-1)dx$
$= \int x\sec^2(x)dx - \int xdx$ Using integration by parts where u=x v=tan($x$)
$du = dxdv = \sec^2(x)dx$ yields:

$$= x\tan(x) - \int \tan(x)dx - \int xdx$$

Both integrals are standard:

$$= x\tan(x) + \ln|\cos(x)| - \frac{1}{2}x^2 + C$$

## Problem 8

$$\int x^2 \cos^2(x)\, dx$$

Rewriting the integral: $=\int x^2\left(\frac{1+\cos(2x)}{2}\right)dx$
$= \frac{1}{2}\int x^2(1+\cos(2x))dx$
$= \frac{1}{2}\int x^2dx + \frac{1}{2}\int x^2\cos(2x)dx$ Using integration by parts where

| Sign | D | I |
|---|---|---|
| + | $x^2$ | $\cos(2x)$ |
| − | $2x$ | $\frac{1}{2}\sin(2x)$ |
| + | 2 | $-\frac{1}{4}\cos(2x)$ |
| − | 0 | $-\frac{1}{8}\sin(2x)$ |

yields: $=1\frac{}{6x^3+\frac{1}{2}\left(\frac{1}{2}x^2\sin(2x)+\frac{1}{2}x\cos(2x)-\frac{1}{4}\sin(2x)\right)+C=\frac{1}{6}x^3+\frac{1}{4}x^2\sin(2x)+\frac{1}{4}x\cos(2x)-\frac{1}{8}\sin(2x)+C}$

## Problem 9

$$\int x^5\sqrt{2-x^3}dx$$

Rewriting the integral:

$$= \int x^2\cdot x^3\sqrt{2-x^3}dx$$

Substituting u=2-x$^3$
$2-u = x^3$
$-du = 3x^2dx$ yields: $=-1\frac{}{3\int(2-u)\sqrt{u}du=-\frac{1}{3}\int\left(2\sqrt{u}-u^{3/2}\right)du=-\frac{2}{3}\int\sqrt{u}du+\frac{1}{3}\int u^{3/2}du}$ Both integrals are standard:

$$= -\frac{4}{9}u^{3/2} + \frac{2}{15}u^{5/2} + C$$

Undoing the substitution(s):

$$= -\frac{4}{9}(2-x^3)^{3/2} + \frac{2}{15}(2-x^3)^{5/2} + C$$

## Problem 10

$$\int \frac{1}{\sqrt{x^2+4}}dx$$

This integral is standard:

$$= \ln\left|x+\sqrt{x^2+4}\right| + C$$

## Problem 11

$$\int \frac{x^2}{\sqrt{25+x^2}}dx$$

Rewriting the integral: $=\int \frac{x^2}{\sqrt{25\left(1+\left(\frac{x}{5}\right)^2\right)}}dx$
$= \frac{1}{5}\int \frac{x^2}{\sqrt{1+(\frac{x}{5})^2}}dx$ Substituting $\tan\theta = \frac{x}{5}$
$5\tan\theta = x$
$5\sec^2\theta d\theta = dx$ yields: $=\int \frac{(5\tan\theta)^2}{\sqrt{1+\tan^2\theta}}\sec^2\theta d\theta$
$= 25\int \frac{\tan^2\theta\sec^2\theta}{\sqrt{\sec^2\theta}}d\theta$
$= 25\int(\sec^2\theta-1)\sec\theta d\theta$
$= 25\int(\sec^3\theta-\sec\theta)d\theta$
$= 25\int\sec^3\theta d\theta - 25\int\sec\theta d\theta$ Both integrals are standard: $=25\left(\frac{1}{2}\sec\theta\tan\theta+\frac{1}{2}\ln|\sec\theta+\tan\theta|\right)-$
$25\ln|\sec\theta+\tan\theta|+C$
$= \frac{25}{2}\sec\theta\tan\theta+\frac{25}{2}\ln|\sec\theta+\tan\theta|-25\ln|\sec\theta+\tan\theta|+C$
$= \frac{25}{2}\sec\theta\tan\theta-\frac{25}{2}\ln|\sec\theta+\tan\theta|+C$ Undoing the substitution(s): $=25\frac{}{2\cdot\frac{\sqrt{x^2+25}}{5}\cdot\frac{x}{5}-\frac{25}{2}\ln\left|\frac{\sqrt{x^2+25}}{5}+\frac{x}{5}\right|+C=\frac{1}{2}x\sqrt{x^2+25}-\frac{25}{2}\ln\left|\sqrt{}\right.}$

## Problem 12

$$\int \cos(x)\sqrt{4-\sin^2(x)}dx$$

Substituting u=sin($x$)
$du = \cos(x)dx$ yields:

$$= \int \sqrt{4-u^2}du$$

This integral is standard:

$$= 2\arcsin\left(\frac{u}{2}\right)+\frac{1}{2}u\sqrt{4-u^2}+C$$

Undoing the substitution(s):

$$= 2\arcsin\left(\frac{\sin(x)}{2}\right)+\frac{1}{2}\sin(x)\sqrt{4-\sin^2(x)}+C$$

## Problem 13

$$\int \frac{1}{x^2 - x + 1} dx$$

Rewriting the integral: $= \int \frac{1}{x^2 - x + \frac{1}{4} + \frac{3}{4}} dx$
$= 4 \int \frac{1}{4x^2 - 4x + 1 + 3} dx$
$= 4 \int \frac{1}{(2x-1)^2 + 3}$ Substituting u=2x-1
du=2dx yields:

$$= 2 \int \frac{1}{u^2 + 3} du$$

This integral is standard:

$$= \frac{2}{\sqrt{3}} \arctan\left(\frac{u}{\sqrt{3}}\right) + C$$

Undoing the substitution(s):

$$= \frac{2}{\sqrt{3}} \arctan\left(\frac{2x - 1}{\sqrt{3}}\right) + C$$

## Problem 14

$$\int \sqrt{x^2 + x + 1} dx$$

Rewriting the integral: $= \int \sqrt{x^2 + x + \frac{1}{4} + \frac{3}{4}} dx$
$= \frac{1}{2} \int \sqrt{4x^2 + 4x + 1 + 3} dx$
$= \frac{1}{2} \int \sqrt{(2x+1)^2 + 3} dx$ Substituting: u=2x+1
du=2dx yields:

$$= \frac{1}{4} \int \sqrt{u^2 + 3} du$$

This integral is standard: $= 1 \frac{}{4\left(\frac{1}{2} u \sqrt{u^2+3} + \frac{3}{2} \ln\left|\sqrt{u^2+3} + u\right|\right) + C = \frac{1}{8} u \sqrt{u^2+3} + \frac{3}{8} \ln\left|\sqrt{u^2+3} + u\right| + C}$ Undoing the substitution(s):

$$= \frac{2x + 1}{8} \sqrt{4x^2 + 4x + 4} + \frac{3}{8} \ln\left|\sqrt{4x^2 + 4x + 4} + 2x + 1\right| + C$$

## Problem 15

$$\int \frac{5x + 31}{3x^2 - 4x + 11} dx$$

Rewriting the integral: $= \int \frac{5x - \frac{10}{3} + \frac{103}{3}}{3x^2 - 4x + 11} dx$
$= \frac{5}{6} \int \frac{6x - 4}{3x^2 - 4x + 11} dx + \frac{103}{9} \int \frac{1}{x^2 - \frac{4}{3}x + \frac{11}{3}} dx$
$= \frac{5}{6} \int \frac{6x - 4}{3x^2 - 4x + 11} dx + \frac{103}{9} \int \frac{1}{x^2 - \frac{4}{3}x + \frac{4}{9} + \frac{29}{9}} dx$
$= \frac{5}{6} \int \frac{6x - 4}{3x^2 - 4x + 11} dx + 103 \int \frac{1}{9x^2 - 12x + 4 + 29} dx$

$= \frac{5}{6}\int \frac{6x-4}{3x^2-4x+11}dx + 103\int \frac{1}{(3x-2)^2+29}$ Substituting: u=$3x^2 - 4x + 11v = 3x - 2$
$du = (6x - 4)dxdv = 3dx$ yields:

$$= \frac{5}{6}\int \frac{1}{u} + \frac{103}{3}\int \frac{1}{v^2+29}dv$$

Both integrals are standard:

$$= \frac{5}{6}\ln|u| + \frac{103}{3\sqrt{29}}\arctan\left(\frac{v}{\sqrt{29}}\right) + C$$

Undoing the substitution(s):

$$= \frac{5}{6}\ln\left|3x^2 - 4x + 11\right| + \frac{103}{3\sqrt{29}}\arctan\left(\frac{3x-2}{\sqrt{29}}\right) + C$$

## Problem 16

$$\int \frac{x^4+1}{x^2+2}dx$$

Rewriting the integral: $=\int \frac{x^4+2x^2-2x^2-4+5}{x^2+2}dx$
$= \int \frac{x^2(x^2+2)-2(x^2+2)+5}{x^2+2}dx$
$= \int \left(x^2 - 2 + \frac{5}{x^2+2}\right)dx$
$= \int x^2dx - 2\int dx + 5\int \frac{1}{x^2+2}dx$ All three integrals are standard:

$$= \frac{1}{3}x^3 - 2x + \frac{5}{\sqrt{2}}\arctan\left(\frac{x}{\sqrt{2}}\right) + C$$

## Problem 17

$$\int \frac{1}{5 + 4\cos(x)}dx$$

Substituting u=$\tan\left(\frac{x}{2}\right)$
$du = \frac{1}{2}\sec^2\left(\frac{x}{2}\right)dx$
$du = \frac{1+u^2}{2}dx$
$\frac{2}{1+u^2}du = dx$ yields: $=2\int \frac{1}{5+4\left(\frac{1-u^2}{1+u^2}\right)} \cdot \frac{1}{1+u^2}du$
$= 2\int \frac{1}{5(1+u^2)+4(1-u^2)}du$
$= 2\int \frac{1}{5+5u^2+4-4u^2}du$
$= 2\int \frac{1}{9+u^2}du$ This integral is standard:

$$= \frac{2}{3}\arctan\left(\frac{u}{3}\right) + C$$

Undoing the substitution(s):

$$= \frac{2}{3}\arctan\left(\frac{\tan(\frac{x}{2})}{3}\right) + C$$

## Problem 18

$$\int \frac{\sqrt{x}}{1+x}dx$$

Substituting $\tan\theta = \sqrt{x}$
$\tan^2\theta = x$
$2\tan\theta\sec^2\theta d\theta = dx$ yields: $=2\int \frac{\tan\theta}{1+\tan^2\theta}\tan\theta\sec^2\theta d\theta$
$= 2\int \frac{\tan^2\theta\sec^2\theta}{\sec^2\theta}d\theta$
$= 2\int \left(\sec^2\theta - 1\right)d\theta$
$= 2\int \sec^2\theta - 2\int d\theta$ Both integrals are standard:

$$= 2\tan\theta - 2\theta + C$$

Undoing the substitution(s):

$$= 2\sqrt{x} - 2\arctan\left(\sqrt{x}\right) + C$$

## Problem 19

$$\int \frac{\cos(x)}{4-\sin^2(x)}dx$$

Substituting u=sin($x$)
$du = \cos(x)dx$ yields:

$$= \int \frac{1}{4-u^2}du$$

This integral is standard:

$$= \frac{1}{4}\ln\left|\frac{2+u}{2-u}\right| + C$$

Undoing the substitution(s):

$$= \frac{1}{4}\ln\left|\frac{2+\sin(x)}{2-\sin(x)}\right| + C$$

## Problem 20

$$\int \frac{\cos(2x)}{\cos(x)}dx$$

Rewriting the integral: $=\int \frac{2\cos^2(x)-1}{\cos(x)}dx$
$= \int(2\cos(x) - \sec(x))dx$
$= 2\int\cos(x)dx - \int\sec(x)dx$ Both integrals are standard:

$$= 2\sin(x) - \ln|\sec(x) + \tan(x)| + C$$

## Problem 21

$$\int \frac{\tan(x)}{\ln(\cos(x))} dx$$

Substituting u=-ln(cos($x$))
$du = \tan(x)dx$ yields:

$$= -\int \frac{1}{u} du$$

This integral is standard:

$$= -\ln|u| + C$$

Undoing the substitution(s):

$$= -\ln|-\ln(\cos(x))| + C$$

## Problem 22

$$\int \frac{x^7}{\sqrt{1-x^4}} dx$$

Rewriting the integral:

$$= \int \frac{x^3 \cdot x^4}{\sqrt{1-x^4}} dx$$

Substituting: u=1-x$^4$
$1 - u = x^4$
$-du = 4x^3 dx$ Yields: $=-1\frac{}{4\int \frac{1-u}{\sqrt{u}} du = -\frac{1}{4}\int (u^{-1/2} - \sqrt{u}) du = -\frac{1}{4}\int u^{-1/2} du + \frac{1}{4}\int \sqrt{u} du}$ Both integrals are standard:

$$= -\frac{1}{2}\sqrt{u} + \frac{1}{6}u^{3/2} + C$$

Undoing the substitution(s):

$$= -\frac{1}{2}\sqrt{1-x^4} + \frac{1}{6}(1-x^4)^{3/2} + C$$

## Problem 23

$$\int \ln(1+x)\, dx$$

Substituting u=1+x
du=dx Yields:

$$= \int \ln(u) du$$

This integral is standard:

$$= u\ln(u) - u + C$$

Undoing the substitution(s):

$$= (1+x)\ln(1+x) - (1+x) + C$$

## Problem 24

$$\int x\,(x)\,dx$$

Using integration by parts where u=(x) v=1$\frac{}{2x^2 du=\frac{1}{x\sqrt{x^2-1}}dx dv=x dx}$ yields:

$$= \frac{1}{2}x^2(x) - \frac{1}{2}\int \frac{x}{\sqrt{x^2-1}}dx$$

Substituting w=x$^2 - 1$
$dw = 2xdx$ yields:

$$= -\frac{1}{4}\int \frac{1}{\sqrt{w}}dw$$

This integral is standard:

$$= \frac{1}{2}x^2(x) - \frac{1}{2}\sqrt{w} + C$$

Undoing the substitution(s):

$$= \frac{1}{2}x^2(x) - \frac{1}{2}\sqrt{x^2-1} + C$$

## Problem 25

$$\int \sqrt{x^2+9}dx$$

This integral is standard:

$$= \frac{1}{2}x\sqrt{x^2+9} + \frac{9}{2}\ln\left|\sqrt{x^2+9} + x\right| + C$$

## Problem 26

$$\int \frac{x^2}{\sqrt{4-x^2}}dx$$

Rewriting the integral: $=\int \frac{x^2}{\sqrt{4(1-\frac{x^2}{4})}}dx$
$= \frac{1}{2}\int \frac{x^2}{\sqrt{1-(\frac{x}{2})^2}}dx$ Substituting $\sin\theta = \frac{x}{2}$
$2\sin\theta = x$
$2\cos\theta d\theta = dx$ yields: $=\int \frac{(2\sin\theta)^2\cos\theta}{\sqrt{1-\sin^2\theta}}d\theta$
$= 4\int \frac{\sin^2\theta\cos\theta}{\sqrt{\cos^2\theta}}d\theta$
$= 4\int \sin^2\theta d\theta$
$= 4\int \left(\frac{1-\cos 2\theta}{2}\right)d\theta$
$= 2\int(1-\cos 2\theta)d\theta$
$= 2\int d\theta - 2\int \cos 2\theta d\theta$ Substituting u=$2\theta$
$du = 2d\theta$ yields:

$$= 2\int d\theta - \int \cos(u)du$$

Both integrals are standard:

$$= 2\theta - \sin(u) + C$$

Undoing the substitution(s): $= 2\theta - \sin(2\theta) + C$
$= 2\theta - 2\sin\theta\cos\theta + C$
$= 2\arcsin\left(\frac{x}{2}\right) - 2 \cdot \frac{x}{2} \cdot \frac{\sqrt{4-x^2}}{2} + C$
$= 2\arcsin\left(\frac{x}{2}\right) - \frac{1}{2}x\sqrt{4-x^2} + C$

## Problem 27

$$\int \sqrt{2x - x^2}dx$$

Rewriting the integral: $= \int \sqrt{1 - 1 + 2x - x^2}dx$
$= \int \sqrt{1 - (1 - 2x + x^2)}dx$
$= \int \sqrt{1 - (1-x)^2}dx$ Substituting: u=1-x
du=-dx yields:

$$= -\int \sqrt{1 - u^2}du$$

This integral is standard:

$$= -\frac{1}{2}\arcsin(u) - \frac{1}{2}u\sqrt{1-u^2} + C$$

Undoing the substitution(s):

$$= -\frac{1}{2}\arcsin(1-x) - \frac{1}{2}(1-x)\sqrt{2x-x^2} + C$$

## Problem 28

$$\int \frac{4x-2}{x^3 - x}dx$$

Factoring the integral:

$$= \int \frac{4x-2}{x(x-1)(x+1)}dx$$

Using partial fraction decomposition

$$= \int \left(\frac{A}{x} + \frac{B}{x-1} + \frac{C}{x+1}\right)dx$$

where $A, B, C \in R$ yields: $= \int \left(\frac{2}{x} + \frac{1}{x-1} + \frac{-3}{x+1}\right)dx$
$= 2\int \frac{1}{x}dx + \int \frac{1}{x-1}dx - 3\int \frac{1}{x+1}dx$ Substituting u=x-1 v=x+1
du=dx dv=dx yields:

$$2\int \frac{1}{x}dx + \int \frac{1}{u}du - 3\int \frac{1}{v}dv$$

All integrals are standard: $= 2\ln|x| + \ln|u| - 3\ln|v| + C$
$= \ln\left|\frac{x^2u}{v^3}\right| + C$ Undoing the substitution(s):

$$= \ln\left|\frac{x^2(x-1)}{(x+1)^3}\right| + C$$

## Problem 29

$$\int \frac{x^4}{x^2-2}dx$$

Rewriting the integral: $=\int \frac{x^4-2x^2+2x^2-4+4}{x^2-2}dx$
$= \int \left(\frac{x^4-2x^2}{x^2-2} + \frac{2x^2-4}{x^2-2} + \frac{4}{x^2-2}\right)dx$
$= \int \frac{x^2(x^2-2)}{x^2-2}dx + \int \frac{2(x^2-2)}{x^2-2}dx + 4\int \frac{1}{x^2-2}dx$
$= \int x^2dx + 2\int dx + 4\int \frac{1}{x^2-2}dx$ All three integrals are standard:

$$= \frac{1}{3}x^3 + 2x + \sqrt{2}\ln\left|\frac{x-\sqrt{2}}{x+\sqrt{2}}\right| + C$$

## Problem 30

$$\int \frac{\sec(x)\tan(x)}{\sec(x)+\sec^2(x)}dx$$

Rewriting the integral: $=\int \frac{\tan(x)}{1+\sec(x)} \cdot \frac{\cos(x)}{\cos(x)}dx$
$= \int \frac{\sin(x)}{\cos(x)+1}dx$ Substituting u=$\cos(x)+1$
$du = -\sin(x)dx$ yields:

$$= -\int \frac{1}{u}du$$

This integral is standard:

$$= -\ln|u| + C$$

Undoing the substitution(s):

$$= -\ln|\cos(x)+1| + C$$

## Problem 31

$$\int \frac{x}{(x^2+2x+2)^2}dx$$

Rewriting the integral: $=\int \frac{x}{(x^2+2x+1+1)^2}dx$
$= \int \frac{x}{((x+1)^2+1)^2}dx$ Substituting $\tan\theta = x+1$
$\tan\theta - 1 = x$
$\sec^2\theta d\theta = dx$ yields: $=\int \frac{\tan\theta-1}{(\tan^2\theta+1)^2}\sec^2\theta d\theta$
$= \int \frac{\tan\theta-1}{\sec^2\theta} \cdot \frac{\cos^2\theta}{\cos^2\theta}d\theta$
$= \int \left(\sin\theta\cos\theta - \cos^2\theta\right)$
$= \int \left(\sin\theta\cos\theta - \frac{1+\cos 2\theta}{2}\right)d\theta$
$= \int \sin\theta\cos\theta d\theta - \frac{1}{2}\int d\theta - \frac{1}{2}\int \cos 2\theta d\theta$ Substituting: u=$\sin\theta v = 2\theta$
$du = \cos\theta d\theta dv = 2d\theta$ yields:

$$= \int udu - \frac{1}{2}\int d\theta - \frac{1}{4}\int \cos(u)du$$

All three integrals are standard:

$$= \frac{1}{2}u^2 - \frac{1}{2}\theta - \frac{1}{4}\sin(u) + C$$

Undoing the substitution(s): $=1\frac{}{2\sin^2\theta-\frac{1}{2}\theta-\frac{1}{4}\sin 2\theta+C=\frac{1}{2}\sin^2\theta-\frac{1}{2}\theta-\frac{1}{2}\sin\theta\cos\theta+C=\frac{1}{2}\left(\frac{x+1}{\sqrt{x^2+2x+2}}\right)^2-\frac{1}{2}\arctan(x+1)-\frac{1}{2}\cdot\frac{x+1}{\sqrt{x^2+2x+2}}\cdot\sqrt{}$

## Problem 32

$$\int \frac{x^{1/3}}{x^{1/2}+x^{1/4}}dx$$

Substituting $u^3 = \sqrt[4]{x}$
$u^4 = \sqrt[3]{x}$
$u^6 = \sqrt{x}$
$u^{12} = x$
$12u^{11}du = dx$ yields: $=\int \frac{u^4}{u^6+u^3}12u^{11}du$
$= 12\int \frac{u^{12}}{u^3+1}du$ Performing long division on the integral yields: $=12\int\left(u^9 - u^6 + u^3 - 1 + \frac{1}{u^3+1}\right)du$
$= 12\int u^9du - 12\int u^6du + 12\int u^3du - 12\int du + 12\int \frac{1}{u^3+1}du$ Factoring the last integral:

$$= 12\int \frac{1}{(u+1)(u^2-u+1)}du$$

Using partial fraction decomposition

$$= 12\int\left(\frac{A}{u+1} + \frac{Bu+C}{u^2-u+1}\right)du$$

where $A, B, C \in R$ yields: $=12\int\left(\frac{\frac{1}{3}}{u+1} + \frac{-\frac{1}{3}u+\frac{2}{3}}{u^2-u+1}\right)du$
$= 4\int \frac{1}{u+1}du - 4\int \frac{u-2}{u^2-u+1}du$ Rewriting the second integral: $=4\int \frac{1}{u+1}du - 4\int \frac{u-\frac{1}{2}-\frac{3}{2}}{u^2-u+1}du$
$= 4\int \frac{1}{u+1}du - 4\int \frac{u-\frac{1}{2}}{u^2-u+1}du + 4\int \frac{\frac{3}{2}}{u^2-u+1}du$
$= 4\int \frac{1}{u+1}du - 2\int \frac{2u-1}{u^2-u+1}du + 6\int \frac{1}{u^2-u+\frac{1}{4}+\frac{3}{4}}du$
$= 4\int \frac{1}{u+1}du - 2\int \frac{2u-1}{u^2-u+1}du + 24\int \frac{1}{4u^2-4u+1+3}du$
$= 4\int \frac{1}{u+1}du - 2\int \frac{2u-1}{u^2-u+1}du + 24\int \frac{1}{(2u-1)^2+3}du$ Substituting v=u+1 w=$u^2 - u + 1y = 2u - 1$
$dv = dudw = (2u-1)dudy = 2du$ yields:

$$= 4\int \frac{1}{v}dv - 2\int \frac{1}{w}dw + 12\int \frac{1}{y^2+3}dy$$

All three integrals are standard: $=4\ln|v| - 2\ln|w| + 4\sqrt{3}\arctan\left(\frac{y}{\sqrt{3}}\right) + C$
$= \ln\left|v^4\right| - \ln\left|w^2\right| + 4\sqrt{3}\arctan\left(\frac{y}{\sqrt{3}}\right) + C$
$= \ln\left|\frac{v^4}{w^2}\right| + 4\sqrt{3}\arctan\left(\frac{y}{\sqrt{3}}\right) + C$ The previous four integrals are also standard:

$$= \frac{6}{5}u^{10} - \frac{12}{7}u^7 + 3u^4 - 12u + \ln\left|\frac{v^4}{w^2}\right| + 4\sqrt{3}\arctan\left(\frac{y}{\sqrt{3}}\right) + C$$

Undoing the substitution(s): $=6\frac{}{5u^{10}-\frac{12}{7}u^7+3u^4-12u+\ln\left|\frac{(u+1)^4}{(u^2-u+1)^2}\right|+4\sqrt{3}\arctan\left(\frac{2u-1}{\sqrt{3}}\right)+C=\frac{6}{5}x^{5/6}-\frac{12}{7}x^{7/12}+3x^{1/3}-12x^{1/12}+\cdots+\ln\left|\frac{}{(x}\right.$

## Problem 33

$$\int \frac{1}{1+\cos(2x)}dx$$

Rewriting the integral: $=\int \frac{1}{1+2\cos^2(x)-1}dx$
$= \frac{1}{2}\int \sec^2(x)dx$ This integral is standard:

$$= \frac{1}{2}\tan(x) + C$$

## Problem 34

$$\int \frac{\sec(x)}{\tan(x)}dx$$

Rewriting the integral: $=\int \frac{\sec(x)}{\tan(x)} \cdot \frac{\cos(x)}{\cos(x)}dx$
$= \int \frac{1}{\sin(x)}dx$
$= \int \csc(x)dx$ This integral is standard:

$$= -\ln|\cot(x) + \csc(x)| + C$$

## Problem 35

$$\int \sec^3(x)\tan^3(x)\,dx$$

Rewriting the integral: $=\int \sec^2(x)\tan^2(x)\sec(x)\tan(x)dx$
$= \int \sec^2(x)(\sec^2(x)-1)\sec(x)\tan(x)$
$= \int (\sec^4(x) - \sec^2(x))\sec(x)\tan(x)$ Substituting u=sec(x)
$du = \sec(x)\tan(x)dx$ yields: $=\int \left(u^4 - u^2\right)du$
$= \int u^4 du - \int u^2 du$ Both integrals are standard:

$$= \frac{1}{5}u^5 - \frac{1}{3}u^3 + C$$

Undoing the substitution(s):

$$= \frac{1}{5}\sec^5(x) - \frac{1}{3}\sec^3(x) + C$$

## Problem 36

$$\int x^2 \arctan(x)\,dx$$

Using integration by parts where u=arctan(x)$v = \frac{1}{3}x^3$
$du = \frac{1}{x^2+1}dx dv = x^2 dx$ yields:

$$= \frac{1}{3}x^3\arctan(x) - \frac{1}{3}\int \frac{x^3}{x^2+1}dx$$

## Problem 37

$$\int x \ln^3(x)\, dx$$

Substituting u=ln($x$)
$e^u = x$
$e^u du = dx$ yields:

$$= \int u^3 e^{2u} du$$

Using integration by parts where

| Sign | D | I |
|---|---|---|
| + | $u^3$ | $e^{2u}$ |
| − | $3u^2$ | $\frac{1}{2}e^{2u}$ |
| + | $6u$ | $\frac{1}{4}e^{2u}$ |
| − | 6 | $\frac{1}{8}e^{2u}$ |
| + | 0 | $\frac{1}{16}e^{2u}$ |

yields:

$$= \frac{1}{2}u^3 e^{2u} - \frac{3}{4}u^2 e^{2u} + \frac{3}{4}ue^{2u} - \frac{3}{8}e^{2u} + C$$

Undoing the substitution(s):

$$= \frac{1}{2}x^2 \ln^3(x) - \frac{3}{4}x^2 \ln^2(x) + \frac{3}{4}x^2 \ln(x) - \frac{3}{8}x^2 + C$$

## Problem 38

$$\int \frac{1}{x\sqrt{1+x^2}} dx$$

Substituting $\tan\theta = x$
$\sec^2\theta d\theta dx$ yields: $= \int \frac{\sec^2\theta}{\tan\theta\sqrt{1+\tan^2\theta}} d\theta$
$= \int \frac{\sec^2\theta}{\tan\theta\sqrt{\sec^2\theta}} d\theta$
$= \int \frac{\sec\theta}{\tan\theta} d\theta$
$= \int \frac{\sec\theta}{\tan\theta} \cdot \frac{\cos\theta}{\cos\theta} d\theta$
$= \int \frac{1}{\sin\theta} d\theta$
$= \int \csc\theta d\theta$ This integral is standard:

$$= -\ln|\cot\theta + \csc\theta| + C$$

Undoing the substitution(s): $= -\ln\left|\frac{1}{x} + \frac{\sqrt{x^2+1}}{x}\right| + C$
$= -\ln\left|\frac{1+\sqrt{x^2+1}}{x}\right| + C$
$= \ln\left|\frac{x}{1+\sqrt{x^2+1}}\right| + C$

## Problem 39

$$\int e^x \sqrt{1+e^{2x}}dx$$

Substituting $\tan\theta = e^x$
$\sec^2\theta d\theta = e^x dx$ yields: $=\int \sec^2\theta\sqrt{1+\tan^2\theta}d\theta$
$= \int \sec^2\theta\sqrt{\sec^2\theta}d\theta$
$= \int \sec^3\theta d\theta$ This integral is standard:

$$= \frac{1}{2}\sec\theta\tan\theta + \frac{1}{2}\ln|\sec\theta+\tan\theta| + C$$

Undoing the substitution(s): $=1\frac{}{2e^x\sqrt{1+e^{2x}}+\frac{1}{2}\ln\left|\sqrt{1+e^{2x}}+e^x\right|+C}$

## Problem 40

$$\int \frac{x}{4x-x^2}dx$$

Rewriting the integral:

$$= \int \frac{1}{4-x}dx$$

Substituting u=4-x
du=-dx yields:

$$= -\int \frac{1}{u}du$$

This integral is standard:

$$= -\ln|u| + C$$

Undoing the substitution(s):

$$= -\ln|4-x| + C$$

## Problem 41

$$\int \frac{1}{x^3\sqrt{x^2-9}}dx$$

Rewriting the integral: $=\int \frac{1}{x^3\sqrt{9\left(\frac{x^2}{9}-1\right)}}dx$
$= \frac{1}{3}\int \frac{1}{x^3\sqrt{\left(\frac{x}{3}\right)^2-1}}dx$ Substituting $\sec\theta = \frac{x}{3}$
$3\sec\theta = x$
$3\sec\theta\tan\theta d\theta = dx$ yields: $=1\frac{}{3\int \frac{3\sec\theta\tan\theta}{(3\sec\theta)^3\sqrt{\sec^2\theta-1}}d\theta=\frac{1}{27}\int\frac{\tan\theta}{\sec^2\theta\sqrt{\tan^2\theta}}d\theta=\frac{1}{27}\int\cos^2\theta d\theta=\frac{1}{54}\int(1+\cos 2\theta)d\theta=\frac{1}{54}\int d\theta+\frac{1}{54}\int\cos 2\theta d\theta}$
Substituting u=2$\theta$
$du2d\theta$ yields:

$$= \frac{1}{54}\int d\theta + \frac{1}{108}\int \cos(u)du$$

Both integrals are standard:

$$= \frac{1}{54}\theta + \frac{1}{108}\sin(u) + C$$

Undoing the substitution(s): $=1\frac{}{54\theta+\frac{1}{108}\sin 2\theta+C=\frac{1}{54}\theta+\frac{1}{54}\sin\theta\cos\theta+C=\frac{1}{54}\left(\frac{x}{3}\right)+\frac{1}{54}\cdot\frac{\sqrt{x^2-9}}{x}\cdot\frac{3}{x}+C=\frac{1}{54}\left(\frac{x}{3}\right)+\frac{\sqrt{x^2-9}}{18x^2}+C}$

## Problem 42

$$\int \frac{x}{(7x+1)^{17}}dx$$

Substituting u=7x+1 $1\frac{}{7u-\frac{1}{7}=x\frac{1}{7}du=dx}$ yields: $=1\frac{}{7\int\frac{\frac{1}{7}u-\frac{1}{7}}{u^{17}}du=\frac{1}{49}\int\frac{u-1}{u^{17}}du=\frac{1}{49}\int(u^{-16}-u^{-17})du=\frac{1}{49}\int u^{-16}du-\frac{1}{49}\int u^{-17}du}$ Both integrals are standard:

$$= -\frac{1}{735u^{15}} + \frac{1}{784u^{16}} + C$$

Undoing the substitution(s):

$$= -\frac{1}{735(7x+1)^{15}} + \frac{1}{784(7x+1)^{16}} + C$$

## Problem 43

$$\int \frac{4x^2+x+1}{4x^3+x}dx$$

Rewriting the integral: $=\int\frac{4x^2+1+x}{x(4x^2+1)}dx$
$= \int\left(\frac{4x^2+1}{x(4x^2+1)}+\frac{x}{x(4x^2+1)}\right)dx$
$= \int\left(\frac{1}{x}+\frac{1}{4x^2+1}\right)dx$
$= \int\frac{1}{x}dx+\int\frac{1}{4x^2+1}dx$ Substituting u=2x du=2dx yields:

$$= \int\frac{1}{x}dx + \frac{1}{2}\int\frac{1}{u^2+1}du$$

Both integrals are standard:

$$= \ln|x| + \frac{1}{2}\arctan(u) + C$$

Undoing the substitution(s):

$$= \ln|x| + \frac{1}{2}\arctan(2x) + C$$

## Problem 44

$$\int \frac{4x^3-x+1}{x^3+1}dx$$

Performing long division on the integral yields: $=\int \left(4-\frac{x+3}{x^3+1}\right)dx$
$=4\int dx-\int \frac{x+3}{(x+1)(x^2-x+1)}dx$ Using partial fraction decomposition

$$=-\int\left(\frac{A}{x+1}+\frac{Bx+C}{x^2-x+1}\right)$$

where $A,B,C\in R$ yields: $=-\int \left(\frac{\frac{2}{3}}{x+1}+\frac{-\frac{2}{3}x+\frac{7}{3}}{x^2-x+1}\right)dx$
$=-\frac{1}{3}\int\left(\frac{2}{x+1}-\frac{2x-7}{x^2-x+1}\right)dx$
$=-\frac{2}{3}\int\frac{1}{x+1}dx+\frac{1}{3}\int\frac{2x-7}{x^2-x+1}dx$
$=-\frac{2}{3}\int\frac{1}{x+1}dx+\frac{1}{3}\int\frac{2x-1-6}{x^2-x+1}dx$
$=-\frac{2}{3}\int\frac{1}{x+1}dx+\frac{1}{3}\int\left(\frac{2x-1}{x^2-x+1}-\frac{6}{x^2-x+1}\right)dx$
$=-\frac{2}{3}\int\frac{1}{x+1}dx+\frac{1}{3}\int\frac{2x-1}{x^2-x+1}dx-2\int\frac{1}{x^2-x+\frac{1}{4}+\frac{3}{4}}dx$
$=-\frac{2}{3}\int\frac{1}{x+1}dx+\frac{1}{3}\int\frac{2x-1}{x^2-x+1}dx-8\int\frac{1}{4x^2-4x+1+3}dx$
$=-\frac{2}{3}\int\frac{1}{x+1}dx+\frac{1}{3}\int\frac{2x-1}{x^2-x+1}dx-8\int\frac{1}{(2x-1)^2+3}dx$ Substituting u=x+1 v=$x^2-x+1w=2x-1$
$du=dxdv=(2x-1)dxdw=2dx$ yields:

$$=-\frac{2}{3}\int\frac{1}{u}du+\frac{1}{3}\int\frac{1}{v}dv-4\int\frac{1}{w^2+3}dw$$

All three integrals are standard: $=-2\frac{}{3\ln|u|+\frac{1}{3}\ln|v|-\frac{4}{\sqrt{3}}\arctan\left(\frac{w}{\sqrt{3}}\right)+C=-\frac{1}{3}\ln|u^2|+\frac{1}{3}\ln|v|-\frac{4}{\sqrt{3}}\arctan\left(\frac{w}{\sqrt{3}}\right)+C=\frac{1}{3}\ln\left|\frac{v}{u^2}\right|-\frac{4}{\sqrt{3}}\arctan\left(\frac{w}{\sqrt{3}}\right.}$
Undoing the substitution(s):

$$=4x+\frac{1}{3}\ln\left|\frac{x^2-x+1}{(x+1)^2}\right|-\frac{4}{\sqrt{3}}\arctan\left(\frac{2x-1}{\sqrt{3}}\right)+C$$

## Problem 45

$$\int\tan^2(x)\sec(x)\,dx$$

Rewriting the integral: $=\int(\sec^2(x)-1)\sec(x)dx$
$=\int(\sec^3(x)-\sec(x))dx$
$=\int\sec^3(x)dx-\int\sec(x)dx$ Both integrals are standard: $=1\frac{}{2\sec(x)\tan(x)+\frac{1}{2}\ln|\sec(x)+\tan(x)-\ln|\sec(x)+\tan(x)|+C=\frac{1}{2}\sec(x)\tan(x)}$

## Problem 46

$$\int\frac{x^2+2x+2}{(x+1)^3}dx$$

Rewriting the integral: $=\int\frac{x^2+2x+1+1}{(x+1)^3}dx$
$=\int\left(\frac{x^2+2x+1}{(x+1)^3}+\frac{1}{(x+1)^3}\right)dx$
$=\int\left(\frac{(x+1)^2}{(x+1)^3}+\frac{1}{(x+1)^3}\right)dx$

$= \int \frac{1}{x+1}dx + \int \frac{1}{(x+1)^3}dx$ Substituting u=x+1
du=dx Yields:

$$= \int \frac{1}{u}du + \int \frac{1}{u^3}du$$

Both integrals are standard:

$$= \ln|u| - \frac{1}{2u^2} + C$$

Undoing the substitution(s):

$$= \ln|x+1| - \frac{1}{2(x+1)^2} + C$$

## Problem 47

$$\int \frac{x^4 + 2x + 2}{x^5 + x^4}dx$$

Rewriting the integral: $= \int \frac{x^4+2(x+1)}{x^5+x^4}dx$
$= \int \left(\frac{x^4}{x^4(x+1)} + \frac{2(x+1)}{x^4(x+1)}\right)dx$
$= \int \frac{1}{x+1}dx + 2\int \frac{1}{x^4}dx$ Substituting u=x+1
du=dx yields:

$$= \int \frac{1}{u}du + 2\int \frac{1}{x^4}dx$$

Both integrals are standard:

$$= \ln|u| - \frac{2}{3x^3} + C$$

Undoing the substitution(s):

$$= \ln|x+1| - \frac{2}{3x^3} + C$$

## Problem 48

$$\int \frac{8x^2 - 4x + 7}{(x^2+1)(4x+1)}dx$$

Rewriting the integral: $= \int \frac{8x^2+8-4x-1}{(x^2+1)(4x+1)}dx$
$= \int \frac{8(x^2+1)-(4x+1)}{(x^2+1)(4x+1)}$
$= \int \left(\frac{8(x^2+1)}{(x^2+1)(4x+1)} - \frac{4x+1}{(x^2+1)(4x+1)}\right)dx$
$= \int \left(\frac{8}{4x+1} - \frac{1}{x^2+1}\right)dx$
$= 8\int \frac{1}{4x+1} - \int \frac{1}{x^2+1}dx$ Substituting u=4x+1
du=4dx yields:

$$= 2\int \frac{1}{u}du - \int \frac{1}{x^2+1}dx$$

Both integrals are standard:

$$= 2\ln|u| - \arctan(x) + C$$

Undoing the substitution(s):

$$= 2\ln|4x+1| - \arctan(x) + C$$

## Problem 49

$$\int \frac{3x^5 - x^4 + 2x^3 - 12x^2 - 2x + 1}{\left(x^3 - 1\right)^2} dx$$

Factoring the integral: $= \int \frac{3x^5 - x^4 + 2x^3 - 12x^2 - 2x + 1}{((x-1)(x^2+x+1))^2} dx$
$= \int \frac{3x^5 - x^4 + 2x^3 - 12x^2 - 2x + 1}{(x-1)^2(x^2+x+1)^2} dx$ Using partial fraction decomposition

$$= \int \left( \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{Cx + D}{x^2 + x + 1} + \frac{Ex + F}{(x^2 + x + 1)^2} \right) dx$$

where $A, B, C, D, E, F \in R$ yields: $= \int \left( \frac{1}{x-1} - \frac{1}{(x-1)^2} + \frac{2x+1}{x^2+x+1} + \frac{4x+2}{(x^2+x+1)^2} \right) dx$
$= \int \frac{1}{x-1} dx - \int \frac{1}{(x-1)^2} dx + \int \frac{2x+1}{x^2+x+1} dx + 2 \int \frac{2x+1}{(x^2+x+1)^2} dx$ Substituting u=x-1 v=$x^2 + x + 1$
$du = dx dv = (2x + 1)dx$ yields:

$$= \int \frac{1}{u} du - \int \frac{1}{u^2} du + \int \frac{1}{v} dv + 2 \int \frac{1}{v^2} dv$$

All four integrals are standard: $= \ln|u| + \frac{1}{u} + \ln|v| - \frac{2}{v} + C$
$= \ln|uv| + \frac{v}{uv} - \frac{2u}{uv} + C$
$= \ln|uv| + \frac{v-2u}{uv} + C$ Undoing the substitution(s): $= \ln|(x-1)(x^2+x+1)| + \frac{x^2+x+1-2(x-1)}{(x-1)(x^2+x+1)} + C$
$= \ln\left|x^3 - 1\right| + \frac{x^2-x+3}{x^3-1} + C$

## Problem 50

$$\int \frac{x}{x^4 + 4x^2 + 8} dx$$

Rewriting the integral: $= \int \frac{x}{x^4+4x^2+4+4} dx$
$= \int \frac{x}{(x^2+2)^2+4} dx$ Substituting u=$x^2 + 2$
$du = 2x dx$ yields:

$$= \frac{1}{2} \int \frac{1}{u^2 + 4} du$$

This integral is standard:

$$= \frac{1}{4} \arctan\left(\frac{u}{2}\right) + C$$

Undoing the substitution(s):

$$= \frac{1}{4} \arctan\left(\frac{x^2 + 2}{2}\right) + C$$

## Problem 51

$$\int \frac{1}{4 + 5\cos(x)} dx$$

Substituting u=$\tan\left(\frac{x}{2}\right)$
$du = \frac{1}{2} \sec^2\left(\frac{x}{2}\right) dx$

$\frac{2}{1+u^2}du = dx$ yields: $= \int \frac{1}{4+5\left(\frac{1-u^2}{1+u^2}\right)} \cdot \frac{2}{1+u^2}du$
$= 2\int \frac{1}{4(1+u^2)+5(1-u^2)}du$
$= 2\int \frac{1}{4+4u^2+5-5u^2}du$
$= 2\int \frac{1}{9-u^2}du$ This integral is standard:

$$= \frac{1}{3}\ln\left|\frac{3+u}{3-u}\right| + C$$

Undoing the substitution(s):

$$= \frac{1}{3}\ln\left|\frac{3+\tan(\frac{x}{2})}{3-\tan(\frac{x}{2})}\right| + C$$

## Problem 52

$$\int \frac{\left(1+x^{2/3}\right)^{3/2}}{x^{1/3}}dx$$

Substituting u=1+x$^{2/3}$
$du = \frac{2}{3}x^{-1/3}dx$ yields:

$$= \frac{3}{2}\int u^{3/2}du$$

This integral is standard:

$$= \frac{3}{5}u^{5/2} + C$$

Undoing the substitution(s):

$$= \frac{3}{5}(1+x^{2/3})^{5/2} + C$$

## Problem 53

$$\int \frac{\arcsin^2(x)}{\sqrt{1-x^2}}dx$$

Substituting u=arcsin$(x)$
$du = \frac{1}{\sqrt{1-x^2}}dx$ yields:

$$= \int u^2 du$$

This integral is standard:

$$= \frac{1}{3}u^3 + C$$

Undoing the substitution(s):

$$= \frac{1}{3}\arcsin^3(x) + C$$

## Problem 54

$$\int \frac{1}{x^{3/2}\left(1+x^{1/3}\right)}dx$$

Substituting u=$x^{1/6}$
$u^2 = x^{1/3}$
$u^6 = x$
$u^9 = x^{3/2}$
$6u^5du = dx$ yields: $=\int \frac{1}{u^9(1+u^2)}6u^5du$
$= 6\int \frac{1}{u^4(1+u^2)}du$ Using partial fraction decomposition

$$= \int \left(\frac{A}{u} + \frac{B}{u^2} + \frac{C}{u^3} + \frac{D}{u^4} + \frac{Eu+F}{u^2+1}\right)du$$

where $A, B, C, D, E, F \in R$ yields: $=6\int \left(-\frac{1}{u^2} + \frac{1}{u^4} + \frac{1}{u^2+1}\right)du$
$= -6\int \frac{1}{u^2}du + 6\int \frac{1}{u^4}du + 6\int \frac{1}{u^2+!}du$ All three integrals are standard:

$$= \frac{6}{u} - \frac{2}{u^3} + 6\arctan(u) + C$$

Undoing the substitution(s):

$$= \frac{6}{x^{1/6}} - \frac{2}{\sqrt{x}} + 6\arctan\left(x^{1/6}\right) + C$$

## Problem 55

$$\int \tan^3(x)\,dx$$

Rewriting the integral: $=\int \tan^2(x)\tan(x)dx$
$= \int(\sec^2(x)-1)\tan(x)dx$
$= \int(\sec^2(x)\tan(x) - \tan(x))dx$
$= \int \sec^2(x)\tan(x)dx - \int \tan(x)dx$ Substituting u=$\tan(x)$
$du = \sec^2(x)dx$ yields:

$$= \int udu - \int \tan(x)dx$$

Both integrals are standard:

$$= \frac{1}{2}u^2 + \ln|\cos(x)| + C$$

Undoing the substitution(s):

$$= \frac{1}{2}\tan^2(x) + \ln|\cos(x)| + C$$

## Problem 56

$$\int \sin^2(x)\cos^4(x)\,dx$$

Rewriting the integral: $=\int(1-\cos^2(x))\cos^4(x)dx$
$=\int(\cos^4(x)-\cos^6(x))dx$
$=\int\cos^4(x)dx-\int\cos^6(x)dx$ Using the reduction rule for $\cos^n(x)$, it's known that: $\int\cos^4(x)dx=\frac{1}{4}\cos^3(x)\sin(x)+\frac{3}{4}\int\cos^2(x)dx$
$=\frac{1}{4}\cos^3(x)\sin(x)+\frac{3}{4}\int\frac{1+\cos(2x)}{2}dx$
$=\frac{1}{4}\cos^3(x)\sin(x)+\frac{3}{8}\int(1+\cos(2x))dx$
$=\frac{1}{4}\cos^3(x)\sin(x)+\frac{3}{8}\int dx+\frac{3}{8}\int\cos(2x)dx$
$=\frac{1}{4}\cos^3(x)\sin(x)+\frac{3}{8}x+\frac{3}{16}\sin(2x)+C$
$=\frac{1}{4}\cos^3(x)\sin(x)+\frac{3}{8}\sin(x)\cos(x)+\frac{3}{8}x+C$ and

$$\int\cos^6(x)dx=\frac{1}{6}\cos^5(x)\sin(x)+\frac{5}{6}\int\cos^4(x)dx$$

Rewriting the integral: $=\int\cos^4(x)dx-\frac{1}{6}\cos^5(x)\sin(x)-\frac{5}{6}\int\cos^4(x)dx$
$=-\frac{1}{6}\cos^5(x)\sin(x)+\frac{1}{6}\int\cos^4(x)dx$
$=-\frac{1}{6}\cos^5(x)\sin(x)+\frac{1}{6}\left(\frac{1}{4}\cos^3(x)\sin(x)+\frac{3}{8}\sin(x)\cos(x)+\frac{3}{8}x+C\right)$
$=-\frac{1}{6}\cos^5(x)\sin(x)+\frac{1}{24}\cos^3(x)\sin(x)+\frac{1}{16}\sin(x)\cos(x)+\frac{1}{16}x+C$

## Problem 57

$$\int\frac{xe^{x^2}}{1+e^{2x^2}}dx$$

Substituting u=$e^{x^2}$
$du=2xe^{x^2}$ yields:

$$=\frac{1}{2}\int\frac{1}{1+u^2}du$$

This integral is standard:

$$=\frac{1}{2}\arctan(u)+C$$

Undoing the substitution(s):

$$=\frac{1}{2}\arctan\left(e^{x^2}\right)+C$$

## Problem 58

$$\int\frac{\cos^3(x)}{\sqrt{\sin(x)}}dx$$

Rewriting the integral: $=\int\frac{\cos(x)\cos^2(x)}{\sqrt{\sin(x)}}dx$
$=\int\frac{\cos(x)(1-\sin^2(x))}{\sqrt{\sin(x)}}dx$ Substituting u=$\sin(x)$
$du=\cos(x)dx$ yields: $=\int\frac{1-u^2}{\sqrt{u}}du$

$= \int \left(\frac{1}{\sqrt{u}} - u^{3/2}\right) du$
$= \int \frac{1}{\sqrt{u}} du - \int u^{3/2} du$ Both integrals are standard:

$$= 2\sqrt{u} - \frac{2}{5}u^{5/2} + C$$

Undoing the substitution(s):

$$= 2\sqrt{\sin(x)} - \frac{2}{5}\left(\sin(x)\right)^{5/2} + C$$

## Problem 59

$$\int x^3 e^{-x^2} dx$$

Rewriting the integral:

$$= \int x \cdot x^2 e^{-x^2} dx$$

Substituting u=-x$^2$
$-\ u = x^2$
$-\ du = 2xdx$ yields:

$$= \frac{1}{2}\int ue^u du$$

Using integration by parts where

| Sign | D | I |
|---|---|---|
| + | $u$ | $e^u$ |
| − | 1 | $e^u$ |
| + | 0 | $e^u$ |

yields:

$$= \frac{1}{2}ue^u - \frac{1}{2}e^u + C$$

Undoing the substitution(s):

$$= -\frac{1}{2}x^2 e^{-x^2} - \frac{1}{2}e^{-x^2} + C$$

## Problem 60

$$\int \sin\left(\sqrt{x}\right) dx$$

Substituting u=$\sqrt{x}$
$u^2 = x$
$2udu = dx$ yields:

$$= 2\int u\sin(u) du$$

Using integration by parts where

| Sign | D | I |
|---|---|---|
| + | $u$ | $\sin(u)$ |
| − | 1 | $-\cos(u)$ |
| + | 0 | $-\sin(u)$ |

yields:

$$= -2u\cos(u) + 2\sin(u) + C$$

Undoing the substitution(s):

$$= -2\sqrt{x}\cos\left(\sqrt{x}\right) + 2\sin\left(\sqrt{x}\right) + C$$

## Problem 61

$$\int \frac{\arcsin(x)}{x^2}dx$$

Substituting $\sin\theta = x$
$\theta = \sin(x)$
$\cos\theta d\theta = dx$ yields: $=\int \frac{\theta\cos\theta}{\sin^2\theta}d\theta$
$= \int u\cot\theta\csc\theta d\theta$ Using integration by parts where

| Sign | D | I |
|---|---|---|
| + | $\theta$ | $\cot\theta\csc\theta$ |
| − | 1 | $-\csc\theta$ |
| + | 0 | $\ln\lvert\csc\theta + \cot\theta\rvert$ |

yields:

$$= -\theta\csc\theta - \ln\lvert\csc\theta + \cot\theta\rvert + C$$

Undoing the substitution(s): $=$-arcsin$(x)\frac{}{x-\ln\left\lvert\frac{1}{x}+\frac{\sqrt{1-x^2}}{x}\right\rvert+C=-\frac{\arcsin(x)}{x}-\ln\left\lvert\frac{1+\sqrt{1-x^2}}{x}\right\rvert+C$

## Problem 62

$$\int \sqrt{x^2-9}dx$$

This integral is standard:

$$= \frac{1}{2}x\sqrt{x^2-9} - \frac{9}{2}\ln\left\lvert\sqrt{x^2-9}+x\right\rvert + C$$

## Problem 63

$$\int x^2\sqrt{1-x^2}dx$$

Substituting $\sin\theta = x$
$\cos\theta d\theta = dx$ yields: $=\int \sin^2\theta\cos\theta\sqrt{1-\sin^2\theta}d\theta$
$= \int \sin^2\theta\cos\theta\sqrt{\cos^2\theta}d\theta$
$= \int \sin^2\theta\cos^2\theta d\theta$

$= \frac{1}{4}\int 4\sin^2\theta\cos^2\theta d\theta$
$= \frac{1}{4}\int (2\sin\theta\cos\theta)^2 d\theta$
$= \frac{1}{4}\int \sin^2 2\theta d\theta$
$= \frac{1}{4}\int \left(\frac{1-\cos 4\theta}{2}\right) d\theta$
$= \frac{1}{8}\int (1-\cos 4\theta) d\theta$
$= \frac{1}{8}\int d\theta - \frac{1}{8}\int \cos 4\theta d\theta$ Substituting u=4$\theta$
$du = 4d\theta$ yields:

$$= \frac{1}{8}\int d\theta - \frac{1}{32}\int \cos(u) du$$

Both integrals are standard:

$$= \frac{1}{8}\theta - \frac{1}{32}\sin(u) + C$$

Undoing the substitution(s): $=1\frac{}{8\theta-\frac{1}{32}\sin 4\theta+C=\frac{1}{8}\theta-\frac{1}{16}\sin 2\theta\cos 2\theta+C=\frac{1}{8}\theta-\frac{1}{8}\sin\theta\cos\theta(\cos^2\theta-\sin^2\theta)+C=\frac{1}{8}\arcsin(x)-\frac{1}{8}x\sqrt{1-x^2}\left(\left(\sqrt{1-}\right.\right.$

## Problem 64

$$\int x\sqrt{2x-x^2}dx$$

Rewriting the integral: $=\int \left(1-\frac{1}{2}(2-2x)\right)\sqrt{2x-x^2}dx$
$= \int \sqrt{2x-x^2}dx - \frac{1}{2}\int (2-2x)\sqrt{2x-x^2}dx$
$= \int \sqrt{1-1+2x-x^2}dx - \frac{1}{2}\int (2-2x)\sqrt{2x-x^2}dx$
$= \int \sqrt{1-(1-2x+x^2)}dx - \frac{1}{2}\int (2-2x)\sqrt{2x-x^2}dx$
$= \int \sqrt{1-(1-x)^2}dx - \frac{1}{2}\int (2-2x)\sqrt{2x-x^2}dx$ Substituting u=1-x v=2x-$x^2$
$du = -dx dv = (2-2x)dx$

$$= -\int \sqrt{1-u^2}du - \frac{1}{2}\int \sqrt{v}dv$$

Both integrals are standard:

$$= -\frac{1}{2}\arcsin(u) - \frac{1}{2}u\sqrt{1-u^2} - \frac{1}{3}v^{3/2} + C$$

Undoing the substitution(s):

$$= -\frac{1}{2}\arcsin(1-x) - \frac{1-x}{2}\sqrt{2x-x^2} - \frac{1}{3}(2x-x^2)^{3/2} + C$$

## Problem 65

$$\int \frac{x-2}{4x^2+4x+1}dx$$

Rewriting the integral:

$$= \int \frac{x-2}{(2x+1)^2}dx$$

Substituting u=2x+1
u-1$\frac{}{2=x\frac{1}{2}du=dx}$ yields: $=1\frac{}{2\int\frac{\frac{1}{2}u-\frac{5}{2}}{u^2}du=\frac{1}{4}\int\frac{u-5}{u^2}du=\frac{1}{4}\int\left(\frac{1}{u}-\frac{5}{u^2}\right)du=\frac{1}{4}\int\frac{1}{u}du-\frac{5}{4}\int\frac{1}{u^2}du}$ Both integrals are standard:

$$= \frac{1}{4}\ln|u| + \frac{5}{4u} + C$$

Undoing the substitution(s): $=1\frac{}{4\ln|2x+1|+\frac{5}{4(2x+1)}+C=\frac{1}{4}\ln|2x+1|+\frac{5}{8x+4}+C}$

## Problem 66

$$\int \frac{2x^2-5x-1}{x^3-2x^2-x+2}dx$$

Factoring the integral: $=\int\frac{2x^2-5x-1}{x^2(x-2)-(x-2)}dx$
$=\int\frac{2x^2-5x-1}{(x^2-1)(x-2)}dx$
$=\int\frac{2x^2-5x-1}{(x-1)(x+1)(x-2)}dx$ Using partial fraction decomposition

$$= \int\left(\frac{A}{x-1}+\frac{B}{x+1}+\frac{C}{x-2}\right)dx$$

where $A, B, C \in R$ yields: $=\int\left(\frac{2}{x-1}+\frac{1}{x+1}+\frac{-1}{x-2}\right)dx$
$=2\int\frac{1}{x-1}dx+\int\frac{1}{x+1}dx-\int\frac{1}{x-2}dx$ Substituting $u_1 = x-1 u_2 = x+1 u_3 = x-2$
$du_1 = dx du_2 = dx du_3 = dx$ yields:

$$= 2\int\frac{1}{u_1}du_1+\int\frac{1}{u_2}du_2-\int\frac{1}{u_3}du_3$$

All three integrals are standard: $=2\ln|u_1|+\ln|u_2|-\ln|u_3|+C$
$=\ln\left|u_1^2\right|+\ln|u_2|-\ln|u_3|+C$
$=\ln\left|\frac{u_1^2u_2}{u_3}\right|+C$ Undoing the substitution(s):

$$= \ln\left|\frac{(x-1)^2(x+1)}{x-2}\right|+C$$

## Problem 67

$$\int\frac{e^{2x}}{e^{2x}-1}dx$$

Substituting u=$e^{2x}-1$
$du = 2e^{2x}dx$ yields:

$$= \frac{1}{2}\int\frac{1}{u}du$$

This integral is standard:

$$= \frac{1}{2}\ln|u|+C$$

Undoing the substitution(s):

$$= \frac{1}{2}\ln\left|e^{2x}-1\right|+C$$

## Problem 68

$$\int \frac{\cos(x)}{\sin^2(x) - 3\sin(x) + 2} dx$$

Substituting u=sin($x$)
$du = \cos(x)dx$ yields:

$$= \int \frac{1}{u^2 - 3u + 2} du$$

Factoring the integral:

$$= \int \frac{1}{(u-2)(u-1)} du$$

Using partial fraction decomposition

$$= \int \left( \frac{A}{u-2} + \frac{B}{u-1} \right) du$$

where $A, B \in R$ yields: $= \int \left( \frac{1}{u-2} + \frac{-1}{u-1} \right) du$
$= \int \frac{1}{u-2} du - \int \frac{1}{u-1} du$ Substituting $v_1 = u - 2 v_2 = u - 1$
$dv_1 = du dv_2 = du$ yields:

$$= \int \frac{1}{v_1} dv_1 - \int \frac{1}{v_2} dv_2$$

Both integrals are standard: $= \ln|v_1| - \ln|v_2| + C$
$= \ln\left|\frac{v_1}{v_2}\right| + C$ Undoing the substitution(s):

$$= \ln\left|\frac{\sin(x) - 2}{\sin(x) - 1}\right| + C$$

## Problem 69

$$\int \frac{2x^3 + 3x^2 + 4}{(x+1)^4} dx$$

Substituting u=x+1
u-1=x
du=dx yields: $= \int \frac{2(u-1)^3 + 3(u-1)^2 + 4}{u^4} du$
$= \int \frac{2(u^3 - 3u^2 + 3u - 1) + 3(u^2 - 2u + 1) + 4}{u^4} du$
$= \int \frac{2u^3 - 6u^2 + 6u - 2 + 3u^2 - 6u + 3 + 4}{u^4} du$
$= \int \frac{2u^3 - 3u^2 + 5}{u^4} du$
$= \int \left( \frac{2}{u} - \frac{3}{u^2} + \frac{5}{u^4} \right) du$
$= 2\int \frac{1}{u} du - 3\int \frac{1}{u^2} du + 5\int \frac{1}{u^4} du$ All three are standard integrals:

$$= 2\ln|u| + \frac{3}{u} - \frac{5}{3u^3} + C$$

Undoing the substitution(s):

$$= 2\ln|x+1| + \frac{3}{x+1} - \frac{5}{3(x+1)^3} + C$$

## Problem 70

$$\int \frac{\sec^2(x)}{\tan^2(x)+2\tan(x)+2}dx$$

Rewriting the integral: $=\int \frac{\sec^2(x)}{\tan^2(x)+2\tan(x)+1+1}dx$
$= \int \frac{\sec^2(x)}{(\tan(x)+1)^2+1}dx$ Substituting u=$\tan(x)+1$
$du = \sec^2(x)dx$ yields:

$$= \int \frac{1}{u^2+1}du$$

This integral is standard:

$$= \arctan(u)+C$$

Undoing the substitution(s):

$$= \arctan(\tan(x)+1)+C$$

## Problem 71

$$\int \frac{x^3+x^2+2x+1}{x^4+2x^2+1}dx$$

Rewriting the integral: $=\int \frac{x^3+x+x+x^2+1}{(x^2+1)^2}dx$
$= \int \frac{x+x(x^2+1)+x^2+1}{(x^2+1)^2}dx$
$= \int \left(\frac{x}{(x^2+1)^2}+\frac{x(x^2+1)}{(x^2+1)^2}+\frac{x^2+1}{(x^2+1)^2}\right)dx$
$= \int \frac{x}{(x^2+1)^2}dx + \int \frac{x}{x^2+1}dx + \int \frac{1}{x^2+1}dx$ Substituting u=$x^2+1$
$du = 2xdx$ yields:

$$= \frac{1}{2}\int \frac{1}{u^2}du + \frac{1}{2}\int \frac{1}{u}du$$

Both are standard integrals:

$$= -\frac{1}{2u}+\frac{1}{2}\ln|u|+\arctan(x)+C$$

Undoing the substitution(s):

$$= -\frac{1}{2x^2+2}+\frac{1}{2}\ln\left|x^2+1\right|+\arctan(x)+C$$

## Problem 72

$$\int \frac{3+\cos(x)}{2-\cos(x)}dx$$

Rewriting the integral: =-$\int \frac{\cos(x)+3}{\cos(x)-2}dx$
$= -\int \frac{\cos(x)-2+5}{\cos(x)-2}dx$
$= -\int \left(\frac{\cos(x)-2}{\cos(x)-2}+\frac{5}{\cos(x)-2}\right)dx$
$= -\int dx - 5\int \frac{1}{\cos(x)-2}dx$ Substituting u=$\tan\left(\frac{x}{2}\right)$

$du = \frac{1}{2}\sec^2\left(\frac{x}{2}\right)dx$
$\frac{2}{1+u^2}du = dx$ yields: $= -5\int \frac{1}{\frac{1-u^2}{1+u^2}-2} \cdot \frac{2}{1+u^2}du$
$= -10\int \frac{1}{1-u^2-2(1+u^2)}du$
$= -10\int \frac{1}{1-u^2-2-2u^2}du$
$= -10\int \frac{1}{-1-3u^2}du$
$= 10\int \frac{1}{1+3u^2}du$ Both integrals are standard:

$$= -x + \frac{10}{\sqrt{3}}\arctan\left(\sqrt{3}u\right) + C$$

Undoing the substitution(s):

$$= -x + \frac{10}{\sqrt{3}}\arctan\left(\sqrt{3}\tan\left(\frac{x}{2}\right)\right) + C$$

## Problem 73

$$\int x^5\sqrt{x^3-1}dx$$

Rewriting the integral:

$$= \int x^2 \cdot x^3\sqrt{x^3-1}dx$$

Substituting u=$x^3 - 1$
$u + 1 = x^3$
$du = 3x^2dx$ yields: $= 1\frac{}{3\int (u+1)\sqrt{u}du = \frac{1}{3}\int\left(u^{3/2}+\sqrt{u}\right)du = \frac{1}{3}\int u^{3/2}du + \frac{1}{3}\int\sqrt{u}du}$ Both integrals are standard:

$$= \frac{2}{15}u^{5/2} + \frac{2}{9}u^{3/2} + C$$

Undoing the substitution(s):

$$= \frac{2}{15}(x^3-1)^{5/2} + \frac{2}{9}(x^3-1)^{3/2} + C$$

## Problem 74

$$\int \frac{1}{2 + 2\cos(x) + \sin(x)}dx$$

Substituting u=$\tan\left(\frac{x}{2}\right)$
$du = \frac{1}{2}\sec^2\left(\frac{x}{2}\right)dx$
$\frac{2}{1+u^2}du = dx$ yields: $= \int \frac{1}{2+2\left(\frac{1-u^2}{1+u^2}\right)+\frac{2u}{1+u^2}} \cdot \frac{2}{1+u^2}du$
$= 2\int \frac{1}{2(1+u^2)+2(1-u^2)+2u}du$
$= 2\int \frac{1}{2+2u^2+2-2u^2+2u}du$
$= 2\int \frac{1}{4+2u}du$

$= \int \frac{1}{2+u} du$ Substituting v=2+u
dv=du yields:

$$= \int \frac{1}{v} dv$$

This integral is standard:

$$= \ln|v| + C$$

Undoing the substitution(s):

$$= \ln\left|2 + \tan\left(\frac{x}{2}\right)\right| + C$$

## Problem 75

$$\int \frac{\sqrt{1+\sin(x)}}{\sec(x)} dx$$

Rewriting the integral:

$$= \int \cos(x)\sqrt{1+\sin(x)} dx$$

Substituting u=1+sin($x$)
$du = \cos(x)dx$ yields:

$$= \int \sqrt{u} du$$

This integral is standard:

$$= \frac{2}{3}u^{3/2} + C$$

Undoing the substitution(s):

$$= \frac{2}{3}(1+\sin(x))^{3/2} + C$$

## Problem 76

$$\int \frac{1}{x^{2/3}\left(1+x^{2/3}\right)} dx$$

Substituting u=$\sqrt[3]{x}$
$du = \frac{1}{3x^{2/3}} dx$ yields:

$$= 3\int \frac{1}{1+u^2} du$$

This integral is standard:

$$= 3\arctan(u) + C$$

Undoing the substitution(s):

$$= 3\arctan\left(\sqrt[3]{x}\right) + C$$

## Problem 77

$$\int \frac{\sin(x)}{\sin(2x)}dx$$

Rewriting the integral: $=\int \frac{\sin(x)}{2\sin(x)\cos(x)}dx$
$= \frac{1}{2}\int \frac{1}{\cos(x)}dx$
$= \frac{1}{2}\int \sec(x)dx$ This integral is standard:

$$= \frac{1}{2}\ln|\sec(x)+\tan(x)| + C$$

## Problem 78

$$\int \sqrt{1+\cos(x)}dx$$

Rewriting the integral: $=\int \sqrt{1+\cos(x)}\cdot\frac{\sqrt{1-\cos(x)}}{\sqrt{1-\cos(x)}}dx$
$= \int \frac{\sqrt{1-\cos^2(x)}}{\sqrt{1-\cos(x)}}dx$
$= \int \frac{\sqrt{\sin^2(x)}}{\sqrt{1-\cos(x)}}dx$
$= \int \frac{\sin(x)}{\sqrt{1-\cos(x)}}dx$ Substituting u=1-cos($x$)
$du = \sin(x)dx$ yields:

$$= \int \frac{1}{\sqrt{u}}du$$

This integral is standard:

$$= 2\sqrt{u} + C$$

Undoing the substitution(s):

$$= 2\sqrt{1-\cos(x)} + C$$

## Problem 79

$$\int \sqrt{1+\sin(x)}dx$$

Rewriting the integral: $=\int \sqrt{1+\sin(x)}\cdot\frac{\sqrt{1-\sin(x)}}{\sqrt{1-\sin(x)}}dx$
$= \int \frac{1-\sin^2(x)}{\sqrt{1-\sin(x)}}dx$
$= \int \frac{\sqrt{\cos^2(x)}}{\sqrt{1-\sin(x)}}dx$
$= \int \frac{\cos(x)}{\sqrt{1-\sin(x)}}dx$ Substituting u=1-sin($x$)
$du = -\cos(x)dx$ yields:

$$= -\int \frac{1}{\sqrt{u}}du$$

This integral is standard:

$$= -2\sqrt{u} + C$$

## Problem 80

$$\int \frac{\sec^2(x)}{1-\tan^2(x)}dx$$

Substituting u=tan($x$)
$du = \sec^2(x)dx$ yields:

$$= \int \frac{1}{1-u^2}du$$

This integral is standard:

$$= \frac{1}{2}\ln\left|\frac{1+u}{1-u}\right| + C$$

Undoing the substitution(s):

$$= \frac{1}{2}\ln\left|\frac{1+\tan(x)}{1-\tan(x)}\right| + C$$

## Problem 81

$$\int \ln\left(x^2+x+1\right)dx$$

Using integration by parts where u=ln($x^2+x+1$)$v = x$
$du = \frac{2x+1}{x^2+x+1}dxdv = dx$ yields:

$$= x\ln(x^2+x+1) - \int \frac{2x^2+x}{x^2+x+1}dx$$

Performing long division on the integral yields: =-$\int \left(2 + \frac{-x-2}{x^2+x+1}\right)dx$
$= -2\int dx + \int \frac{x+2}{x^2+x+1}dx$
$= -2\int dx + \int \frac{x+\frac{1}{2}+\frac{3}{2}}{x^2+x+1}dx$
$= -2\int dx + \frac{1}{2}\int \frac{2x+1}{x^2+x+1}dx + \frac{3}{2}\int \frac{1}{x^2+x+1}dx$
$= -2\int dx + \frac{1}{2}\int \frac{2x+1}{x^2+x+1}dx + \frac{3}{2}\int \frac{1}{x^2+x+\frac{1}{4}+\frac{3}{4}}dx$
$= -2\int dx + \frac{1}{2}\int \frac{2x+1}{x^2+x+1}dx + 6\int \frac{1}{4x^2+4x+1+3}dx$
$= -2\int dx + \frac{1}{2}\int \frac{2x+1}{x^2+x+1}dx + 6\int \frac{1}{(2x+1)^2+3}dx$ Substituting w=x$^2+x+1z = 2x+1$
$dw = (2x+1)dxdz = 2dx$ yields:

$$= -2\int dx + \frac{1}{2}\int \frac{1}{w}dw + 3\int \frac{1}{z^2+3}$$

All three integrals are standard:

$$= -2x + \frac{1}{2}\ln|w| + \frac{3}{\sqrt{3}}\arctan\left(\frac{z}{\sqrt{3}}\right) + C$$

Undoing the substitution(s):

$$= x\ln(x^2+x+1) - 2x + \frac{1}{2}\ln\left|x^2+x+1\right| + \sqrt{3}\arctan\left(\frac{2x+1}{\sqrt{3}}\right) + C$$

## Problem 82

$$\int e^x \arcsin\left(e^x\right) dx$$

Substituting u=e$^x$
$du = e^x dx$ yields:

$$= \int \arcsin(u) du$$

This integral is standard:

$$= u \arcsin(u) + \sqrt{1-u^2} + C$$

Undoing the substitution(s):

$$= e^x \arcsin\left(e^x\right) + \sqrt{1-e^{2x}} + C$$

## Problem 83

$$\int \frac{\arctan(x)}{x^2} dx$$

Using integration by parts where: u=arctan$(x) v = -\frac{1}{x}$
$du = \frac{1}{1+x^2} dx dv = \frac{1}{x^2} dx$ yields:

$$= -\frac{\arctan(x)}{x} + \int \frac{1}{x(1+x^2)} dx$$

Using partial fraction decomposition

$$= \int \left(\frac{A}{x} + \frac{Bx+C}{1+x^2}\right) dx$$

where $A, B, C \in R$ yields: $=\int \left(\frac{1}{x} - \frac{x}{x^2+1}\right) dx$
$= \int \frac{1}{x} dx - \int \frac{x}{x^2+1} dx$ Substituting u=x$^2+1$
$du = 2x dx$ yields:

$$= -\frac{1}{2} \int \frac{1}{u} du$$

This integral is standard:

$$= -\frac{1}{2} \ln|u| + C$$

Undoing the substitution(s): =-arctan$(x)$ $x + \ln|x| - \frac{1}{2}\ln|x^2+1| + C = -\frac{\arctan(x)}{x} + \ln\left|\frac{x}{\sqrt{x^2+1}}\right| + C$

## Problem 84

$$\int \frac{x^2}{\sqrt{x^2-25}} dx$$

Rewriting the integral: $=\int \frac{x^2}{\sqrt{25\left(\frac{x^2}{25}-1\right)}} dx$
$= \frac{1}{5} \int \frac{x^2}{\sqrt{\left(\frac{x}{5}\right)^2-1}} dx$ Substituting $\sec\theta = \frac{x}{5}$

$5\sec\theta = x$
$5\sec\theta\tan\theta d\theta = dx$ yields: $= \int \frac{(5\sec\theta)^2}{\sqrt{\sec^2\theta - 1}} \sec\theta\tan\theta d\theta$
$= 25\int \frac{\sec^3\theta\tan\theta}{\sqrt{\tan^2\theta}} d\theta$
$= 25\int \sec^3\theta d\theta$ This integral is standard: $= 25\left(\frac{1}{2}\sec\theta\tan\theta + \frac{1}{2}\ln|\sec\theta + \tan\theta| + C\right)$
$= \frac{25}{2}\sec\theta\tan\theta + \frac{25}{2}\ln|\sec\theta + \tan\theta| + C$ Undoing the substitution(s): $= 25\frac{}{2\cdot\frac{x}{5}\cdot\frac{\sqrt{x^2-25}}{5} + \frac{25}{2}\ln\left|\frac{x}{5} + \frac{\sqrt{x^2-25}}{5}\right| + C = \frac{1}{2}x\sqrt{x^2-25} + \frac{25}{2}\ln\left|x\right.}$

## Problem 85

$$\int \frac{x^3}{(x^2+1)^2} dx$$

Rewriting the integral:

$$= \int \frac{x \cdot x^2}{(x^2+1)^2} dx$$

Substituting u=$x^2 + 1$
$u - 1 = x^2$
$du = 2xdx$ yields: $= 1\frac{}{2\int \frac{u-1}{u^2} du = \frac{1}{2}\int\left(\frac{1}{u} - \frac{1}{u^2}\right)du = \frac{1}{2}\int \frac{1}{u} du - \frac{1}{2}\int \frac{1}{u^2} du}$ All three integrals are standard:

$$= \frac{1}{2}\ln|u| - \frac{1}{2u} + C$$

Undoing the substitution(s):

$$= \frac{1}{2}\ln\left|x^2+1\right| + \frac{1}{2x^2+2} + C$$

## Problem 86

$$\int \frac{1}{x\sqrt{6x - x^2}} dx$$

Rewriting the integral: $= \int \frac{1}{x\sqrt{9-9+6x-x^2}} dx$
$= \int \frac{1}{x\sqrt{9-(9-6x+x^2)}} dx$
$= \int \frac{1}{x\sqrt{9-(3-x)^2}} dx$
$= \int \frac{1}{x\sqrt{9\left(1-\frac{(3-x)^2}{9}\right)}} dx$
$= \frac{1}{3}\int \frac{1}{x\sqrt{1-\left(\frac{3-x}{3}\right)^2}} dx$ Substituting $\sin\theta = \frac{3-x}{3}$
$3 - 3\sin\theta = x$
$-3\cos\theta d\theta = dx$ yields: $= -\int \frac{\cos\theta}{(3-3\sin\theta)\sqrt{1-\sin^2\theta}} d\theta$
$= -\int \frac{\cos\theta}{(3-3\sin\theta)\sqrt{\cos^2\theta}} d\theta$
$= -\int \frac{1}{3-3\sin\theta} d\theta$
$= -\frac{1}{3}\int \frac{1}{1-\sin\theta} \cdot \frac{1+\sin\theta}{1+\sin\theta} d\theta$
$= -\frac{1}{3}\int \frac{1+\sin\theta}{1-\sin^2\theta} d\theta$
$= -\frac{1}{3}\int \frac{1+\sin\theta}{\cos^2\theta} d\theta$

$= -\frac{1}{3}\int \left(\sec^2\theta + \sec\theta\tan\theta\right) d\theta$
$= -\frac{1}{3}\int \sec^2\theta d\theta - \frac{1}{3}\int \sec\theta\tan\theta d\theta$ Both integrals are standard:

$$= -\frac{1}{3}\tan\theta - \frac{1}{3}\sec\theta + C$$

Undoing the substitution(s): $=\text{-}1\frac{3-x}{3\cdot\sqrt{6x-x^2}} - \frac{1}{3}\cdot\frac{3}{\sqrt{6x-x^2}} + C = \frac{x-3}{3\sqrt{6x-x^2}} - \frac{3}{3\sqrt{6x-x^2}} + C = \frac{x-6}{3\sqrt{6x-x^2}} + C$

## Problem 87

$$\int \frac{3x+2}{\left(x^2+4\right)^{3/2}} dx$$

Rewriting the integral:

$$\frac{3}{2}\int \frac{2x}{\left(x^2+4\right)^{3/2}} dx + \frac{1}{4}\int \frac{1}{\left(\left(\frac{x}{2}\right)^2+1\right)^{3/2}} dx$$

Substituting $\text{u=x}^2 + 4\tan\theta = \frac{x}{2}$
$du = 2xdx\sec^2\theta d\theta = \frac{1}{2}dx$ yields: $=3\frac{}{2\int \frac{1}{u^{3/2}}du + \frac{1}{2}\int \frac{\sec^2\theta}{(\tan^2\theta+1)^{3/2}}d\theta = \frac{3}{2}\int \frac{1}{u^{3/2}}du + \frac{1}{2}\int \frac{\sec^2\theta}{(\sec^2\theta)^{3/2}}d\theta = \frac{3}{2}\int \frac{1}{u^{3/2}}du + \frac{1}{2}\int \frac{1}{\sec\theta}d\theta = \frac{3}{2}\int \frac{1}{u^{3/2}}$

Both integrals are standard:

$$= -\frac{3}{\sqrt{u}} + \frac{1}{2}\sin\theta + C$$

Undoing the substitution(s): $=\text{-}3\frac{}{\sqrt{x^2+4}} + \frac{1}{2}\cdot\frac{x}{\sqrt{x^2+4}} + C = \frac{x-6}{2\sqrt{x^2+4}} + C$

## Problem 88

$$\int x^{3/2}\ln(x)\, dx$$

Using integration by parts where $\text{u=ln}(x) v = \frac{2}{5}x^{5/2}$
$du = \frac{1}{x}dx dv = x^{3/2}dx$ yields:

$$= \frac{2}{5}x^{5/2}\ln(x) - \frac{2}{5}\int x^{3/2}dx$$

This integral is standard:

$$= \frac{2}{5}x^{5/2}\ln(x) - \frac{4}{25}x^{5/2} + C$$

## Problem 89

$$\int \frac{\sqrt{1+\sin^2(x)}}{\sec(x)\csc(x)} dx$$

Rewriting the integral;

$$= \int \sin(x)\cos(x)\sqrt{1+\sin^2(x)}dx$$

Substituting u=$1+\sin^2(x)$
$du = 2\sin(x)\cos(x)dx$ yields:

$$= \frac{1}{2}\int \sqrt{u}du$$

This integral is standard:

$$= \frac{1}{3}u^{3/2} + C$$

Undoing the substitution(s):

$$= \frac{1}{3}\left(1+\sin^2(x)\right)^{3/2} + C$$

## Problem 90

$$\int \frac{e^{\sqrt{\sin(x)}}}{\sec(x)\sqrt{\sin(x)}}dx$$

Rewriting the integral:

$$= \int \frac{\cos(x)e^{\sqrt{\sin(x)}}}{\sqrt{\sin(x)}}dx$$

Substituting u=$\sqrt{\sin(x)}$
$du = \frac{\cos(x)}{2\sqrt{\sin(x)}}dx$ yields

$$= 2\int e^u du$$

This integral is standard:

$$= 2e^u + C$$

Undoing the substitution(s):

$$= 2e^{\sqrt{\sin(x)}} + C$$

## Problem 91

$$\int xe^x \sin(x)dx$$

Recall that

$$\int e^x \sin(x)dx = \frac{1}{2}e^x \sin(x) - \frac{1}{2}e^x \cos(x)$$

Using integration by parts where

| Sign | D | I |
|---|---|---|
| + | $xe^x$ | $\sin(x)$ |
| − | $e^x + xe^x$ | $-\cos(x)$ |
| + | $2e^x + xe^x$ | $-\sin(x)$ |

yields: =-x$e^x\cos(x) + e^x\sin(x)(1+x) - \int e^x\sin(x)(2+x)dx$
$2\int xe^x\sin(x)dx = -xe^x\cos(x) + e^x\sin(x)(1+x) - 2\int e^x\sin(x)dx$
$= -xe^x\cos(x) + xe^x\sin(x) + e^x\sin(x) - e^x\sin(x) + e^x\cos(x)$
$\int xe^x\sin(x)dx = \frac{1}{2}xe^x(\sin(x)-\cos(x)) + \frac{1}{2}e^x\cos(x) + C$

## Problem 92

$$\int x^2 e^{x^{3/2}} dx$$

Rewriting the integral:

$$= \int x^{3/2} \cdot x^{1/2} e^{x^{3/2}} dx$$

Substituting u=$x^{3/2}$
$du = \frac{3}{2} x^{1/2} dx$ yields:

$$= \frac{2}{3} \int u e^u du$$

Using integration by parts where w=u v=$e^u$
$dw = dudv = e^u du$ yields:

$$= \frac{2}{3} u e^u - \frac{2}{3} \int e^u du$$

This integral is standard:

$$= \frac{2}{3} u e^u - \frac{2}{3} e^u + C$$

Undoing the substitution(s):

$$= \frac{2}{3} x^{3/2} e^{x^{3/2}} - \frac{2}{3} e^{x^{3/2}} + C$$

## Problem 93

$$\int \frac{\arctan(x)}{(x-1)^3} dx$$

Using integration by parts where u=$\arctan(x) v = -\frac{1}{2(x-1)^2} dx$
$du = \frac{1}{1+x^2} dx dv = \frac{1}{(x-1)^3} dx$ yields:

$$= -\frac{\arctan(x)}{2(x-1)^2} + \frac{1}{2} \int \frac{1}{(1+x^2)(x-1)^2} dx$$

Using partial fraction decomposition

$$= \frac{1}{2} \int \left( \frac{Ax+B}{1+x^2} + \frac{C}{(x-1)^2} + \frac{D}{x-1} \right) dx$$

where $A, B, C, D \in R$ yields: $=1\frac{}{2\left(\frac{\frac{1}{2}x+0}{x^2+1}+\frac{-\frac{1}{2}}{x-1}+\frac{\frac{1}{2}}{(x-1)^2}\right)dx=\frac{1}{4}\int\left(\frac{x}{x^2+1}-\frac{1}{x-1}+\frac{1}{(x-1)^2}\right)dx=\frac{1}{4}\int\frac{x}{x^2+1}dx-\frac{1}{4}\int\frac{1}{x-1}dx+\frac{1}{4}\int\frac{1}{(x-1)^2}dx}$
Substituting $w_1 = x^2 + 1 w_2 = x - 1$
$dw_1 = 2x dx dw_2 = dx$ yields: $=1\frac{}{8\int\frac{1}{w_1}dw_1-\frac{1}{4}\int\frac{1}{w_2}dw_2+\frac{1}{4}\int\frac{1}{w_2^2}dw_2}$ All three integrals are standard:
$=1\frac{}{8\ln|w_1|-\frac{1}{4}\ln|w_2|-\frac{1}{4w_2}=\frac{1}{8}\ln\left|\frac{w_1}{w_2^2}\right|-\frac{1}{4w_2}}$ Undoing the substitution(s):

$$= -\frac{\arctan(x)}{2(x-1)^2} + \frac{1}{8} \ln \left| \frac{x^2+1}{(x-1)^2} \right| - \frac{1}{4(x-1)} + C$$

## Problem 94

$$\int \ln\left(1+\sqrt{x}\right)dx$$

Substituting u=1+$\sqrt{x}$
$u-1=\sqrt{x}$
$u^2-2u+1=x$
$(2u-2)du=dx$ yields: $=\int \ln(u)(2u-2)du$
$=2\int u\ln(u)du-2\int \ln(u)du$ Using integration by parts where w=$\ln(u)v=\frac{1}{2}u^2$
$dw=\frac{1}{u}dudv=udu$ yields: $=2\left(\frac{1}{2}u^2\ln(u)-\frac{1}{2}\int udu\right)$
$=u^2\ln(u)-\int udu$ The remaining two integrals are standard: $=\mathrm{u}^2\ln(u)-\frac{1}{2}u^2-2u\ln(u)+2u+C$
$=u\ln(u)\left(u-2\right)-\frac{1}{2}u^2+2u+C$ Undoing the substitution(s):

$$=\left(1+\sqrt{x}\right)\ln\left(1+\sqrt{x}\right)\left(\sqrt{x}-1\right)-\frac{1}{2}\left(1+\sqrt{x}\right)^2+2\sqrt{x}+C$$

## Problem 95

$$\int \frac{2x+3}{\sqrt{3+6x-9x^2}}dx$$

Rewriting the integral: =2x-2$\frac{}{3+\frac{11}{3}}$ $\frac{}{\sqrt{3+6x-9x^2}}dx=-\frac{1}{9}\int\frac{6-18x}{\sqrt{3+6x-9x^2}}+\frac{11}{3}\int\frac{1}{\sqrt{4-1+6x-9x^2}}dx=-\frac{1}{9}\int\frac{6-18x}{\sqrt{3+6x-9x^2}}+\frac{11}{3}\int\frac{1}{\sqrt{4-(1-6x+9x^2)}}dx=$
Substituting u=3+6x-9x$^2v=1-3x$
$du=(6-18x)dxdv=-3dx$ yields:

$$=-\frac{1}{9}\int\frac{1}{\sqrt{u}}du-\frac{11}{9}\int\frac{1}{\sqrt{4-v^2}}dv$$

Both integrals are standard:

$$=-\frac{2}{9}\sqrt{u}-\frac{11}{9}\arcsin\left(\frac{v}{2}\right)+C$$

Undoing the substitution(s):

$$=-\frac{2}{9}\sqrt{3+6x-9x^2}-\frac{11}{9}\arcsin\left(\frac{1-3x}{2}\right)+C$$

## Problem 96

$$\int \frac{1}{2+2\sin(x)+\cos(x)}dx$$

Substituting u=tan$\left(\frac{x}{2}\right)$
$du=\frac{1}{2}\sec^2\left(\frac{x}{2}\right)dx$
$\frac{2}{1+u^2}du=dx$ yields: $=1$ $\frac{}{2+2\left(\frac{2u}{1+u^2}\right)+\left(\frac{1-u^2}{1+u^2}\right)\cdot\frac{2}{1+u^2}du=2\int\frac{1}{2+2u^2+4u+1-u^2}du=2\int\frac{1}{u^2+4u+3}du=2\int\frac{1}{(u+3)(u+1)}du=2\left(\frac{-\frac{1}{2}}{u+3}+\frac{\frac{1}{2}}{u+1}\right)du=-\int\frac{}{u}$
Substituting $\mathrm{v}_1=u+3v_2=u+1$
$dv_1=dudv_2=du$ yields:

$$=-\int\frac{1}{v_1}dv_1+\int\frac{1}{v_2}dv_2$$

Both integrals are standard: =-$\ln|v_1| + \ln|v_2| + C$
$= \ln\left|\frac{v_2}{v_1}\right| + C$ Undoing the substitution(s):

$$\ln\left|\frac{\tan\left(\frac{x}{2}\right)+1}{\tan\left(\frac{x}{2}\right)+3}\right| + C$$

## Problem 97

$$\int \frac{\sin^3(x)}{\cos(x)-1}dx$$

Rewriting the integral: =-$\int \frac{\sin(x)\cdot\sin^2(x)}{1-\cos(x)}dx$
$= -\int \frac{\sin(x)(1-\cos^2(x))}{1-\cos(x)}dx$
$= -\int \frac{\sin(x)(1-\cos(x))(1+\cos(x))}{1-\cos(x)}dx$
$= -\int \sin(x)(1+\cos(x))dx$ Substituting u=1+cos(x)
$du = -\sin(x)dx$ yields:

$$= \int u du$$

This integral is standard:

$$= \frac{1}{2}u^2 + C$$

Undoing the substitution(s):

$$= \frac{1}{2}(1+\cos(x))^2 + C$$

## Problem 98

$$\int x^{3/2}\arctan\left(\sqrt{x}\right)dx$$

Using integration by parts where: u=$\arctan(\sqrt{x})$ $v = \frac{2}{5}x^{5/2}$
$dw = \frac{1}{2\sqrt{x}(1+x)}dx dv = x^{3/2}dx$ yields: =$2\frac{}{5x^{5/2}\arctan(\sqrt{x}) - \frac{1}{5}\int \frac{x^2}{1+x}dx}$ Substituting w=1+x
w-1=x
dw=dx yields: =$-1\frac{}{5\int \frac{(w-1)^2}{w}dw = -\frac{1}{5}\int \frac{w^2-2w+1}{w}dw = -\frac{1}{5}\int\left(w-2+\frac{1}{w}\right)dw = -\frac{1}{5}\int w dw + \frac{2}{5}\int dw - \frac{1}{5}\int \frac{1}{w}dw}$ All three integrals are standard:

$$-\frac{1}{10}w^2 + \frac{2}{5}w - \frac{1}{5}\ln|w| + C$$

Undoing the substitution(s):

$$= \frac{2}{5}x^{5/2}\arctan\left(\sqrt{x}\right) - \frac{1}{10}(1+x)^2 + \frac{2}{5}(1+x) - \frac{1}{5}\ln|1+x| + C$$

## Problem 99

$$\int (\sqrt{x})\, dx$$

Substituting u=$\sqrt{x}$
$u^2 = x$
$2udu = dx$ yields:

$$= 2\int u(u)du$$

Using integration by parts where: w=(u) v=1$\frac{}{2u^2 dw=\frac{1}{u\sqrt{u^2-1}}dudv=udu}$ yields: $=2\left(\frac{1}{2}u^2(u) - \frac{1}{2}\int \frac{u^2}{u\sqrt{u^2-1}}du\right)$
$= u^2(u) - \int \frac{u}{\sqrt{u^2-1}}du$ Substituting z=u$^2 - 1$
$dz = 2udu$ yields:

$$= u^2(u) - \frac{1}{2}\int \frac{1}{\sqrt{z}}dz$$

This integral is standard:

$$= u^2(u) - \sqrt{z} + C$$

Undoing the substitution(s): $=x(\sqrt{x}) - \sqrt{x-1} + C$

## Problem 100

$$\int x\sqrt{\frac{1-x^2}{1+x^2}}dx$$

Substituting $\sin\theta = x^2$
$\cos\theta d\theta = 2xdx$ yields: $=1$

$$2\int \cos\theta\sqrt{\frac{1-\sin\theta}{1+\sin\theta}}d\theta=\frac{1}{2}\int \cos\theta\sqrt{\frac{1-\sin\theta}{1+\sin\theta}\cdot\frac{1-\sin\theta}{1-\sin\theta}}d\theta=\frac{1}{2}\int \cos\theta\sqrt{\frac{(1-\sin\theta)^2}{1-\sin^2\theta}}d\theta=\frac{1}{2}\int \cos\theta\sqrt{\frac{(1-\sin\theta)^2}{\cos^2\theta}}d\theta=\frac{1}{2}\int \cos\theta\cdot\frac{1-}{c}$$

Both integrals are standard:

$$= \frac{1}{2}\theta + \frac{1}{2}\cos\theta + C$$

Undoing the substitution(s):

$$= \frac{1}{2}\arcsin\left(x^2\right) + \frac{1}{2}\sqrt{1-x^4} + C$$

# 4 Advanced Integrals - Proofs

## Advanced Integral 1

$$\int \sec^3(x)dx$$

Rewriting the integral:

$$= \int \sec(x)\sec^2(x)dx$$

Using integration by parts where u=$\sec(x) v = \tan(x)$
$du = \sec(x)\tan(x)dx dv = \sec^2(x)dx$ yields: $= \sec(x)\tan(x) - \int \sec(x)\tan^2(x)dx$
$= \sec(x)\tan(x) - \int \sec(x)(\sec^2(x) - 1)dx$
$= \sec(x)\tan(x) - \int (\sec^3(x) - \sec(x))dx$
$= \sec(x)\tan(x) - \int \sec^3(x)dx + \int \sec(x)dx$
$2\int \sec^3(x)dx = \sec(x)\tan(x) + \ln|\sec(x) + \tan(x)| + C$
$= \frac{1}{2}\sec(x)\tan(x) + \frac{1}{2}\ln|\sec(x) + \tan(x)| + C$

## Advanced Integral 2

$$\int \arcsin(x)dx$$

Using integration by parts where u=$\arcsin(x) v = x$
$du = \frac{1}{\sqrt{1-x^2}}dx dv = dx$ yields:

$$x\arcsin(x) - \int \frac{x}{\sqrt{1-x^2}}dx$$

Substituting u=1-$x^2$
$du = -2xdx$ yields:

$$= \frac{1}{2}\int \frac{1}{\sqrt{u}}du$$

This integral is standard:

$$= \sqrt{u} + C$$

Undoing the substitution(s):

$$= x\arcsin(x) + \sqrt{1-x^2} + C$$

## Advanced Integral 3

$$\int \arccos(x)dx$$

Using integration by parts where u=$\arccos(x) v = x$
$du = -\frac{1}{\sqrt{1-x^2}}dx dv = dx$ yields:

$$x\arccos(x) + \int \frac{x}{\sqrt{1-x^2}}dx$$

Substituting u=1-x$^2$
$du = -2xdx$ yields:

$$= -\frac{1}{2}\int \frac{1}{\sqrt{u}}du$$

This integral is standard:

$$= -\sqrt{u} + C$$

Undoing the substitution(s):

$$= x\arccos(x) - \sqrt{1-x^2} + C$$

## Advanced Integral 4

$$\int \arctan(x)dx$$

Using integration by parts where u=arctan$(x)v = x$
$du = \frac{1}{1+x^2}dxdv = dx$ yields:

$$= x\arctan(x) - \int \frac{x}{1+x^2}dx$$

Substituting u=1+x$^2$
$du = 2xdx$ yields:

$$= -\frac{1}{2}\int \frac{1}{u}du$$

This integral is standard:

$$= x\arctan(x) - \frac{1}{2}\ln\left|1+x^2\right| + C$$

## Advanced Integral 5

$$\int \sqrt{x^2+a^2}dx$$

Rewriting the integral: $=\int \sqrt{a^2\left(\frac{x^2}{a^2}+1\right)}dx$
$= a\int \sqrt{\left(\frac{x}{a}\right)^2+1}dx$ Substituting

## Advanced Integral 6

$$\int \sqrt{x^2-a^2}dx$$

## Advanced Integral 7

$$\int \sqrt{a^2-x^2}dx$$

**Advanced Integral 8**

$$\int \frac{1}{x^2 + a^2} dx$$

**Advanced Integral 9**

$$\int \frac{1}{x^2 - a^2} dx$$

**Advanced Integral 10**

$$\int \frac{1}{a^2 - x^2} dx$$

**Advanced Integral 11**

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx$$

**Advanced Integral 12**

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx$$

**Advanced Integral 13**

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx$$

**Advanced Integral 14**

$$\int \sin^n(x) dx$$

**Advanced Integral 15**

$$\int \cos^n(x) dx$$